

Mehrshad Zand

Backend Software Engineer · .NET

Tehran, Iran · Open to Remote, International

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Military Service: Educational Exemption

PROFESSIONAL SUMMARY

Backend Software Engineer with approximately 5 years of experience in the .NET ecosystem, specializing in scalable, secure enterprise systems. Skilled in Clean Architecture, long-term maintainability, software security, and solving complex business problems across finance, tax, and healthcare platforms. Experienced in reducing technical debt, automating deployment workflows, and building infrastructure that holds up at real-world scale. Seeking remote, international backend engineering opportunities.

TECHNICAL SKILLS

Languages & Frameworks:

C#, .NET 6–10, ASP.NET Core, .NET Framework

Backend & API:

REST APIs, RabbitMQ, YARP, JWT, FluentValidation, MediatR, Dapper, Quartz, xUnit, SignalR, Polly

Architecture:

Clean Architecture, Onion Architecture, Modular Monolith, Microservices, DDD, CQRS, SAGA

Databases:

SQL Server (Advanced), PostgreSQL, Redis, MongoDB, Stored Procedures, TVPs

Security:

RSA/AES Encryption, Access Control (RBAC), OAuth, AFTA Compliance, OWASP

Infrastructure & DevOps:

Docker, .NET Aspire, IIS, PowerShell, Git, Azure DevOps, Linux

UI & Working Knowledge:

Blazor Server/WASM, .NET MAUI, WPF, Spectre.Console (TUI) · Working Knowledge: Java, Python, C++, Go, JS, TS

PROFESSIONAL EXPERIENCE

Backend Software Engineer — Samarayaneh Pishro

Tehran, Iran

2024 – Present

- Designed and maintained scalable backend services using ASP.NET Core for financial and healthcare platforms.
- Built authentication and fine-grained Role-Based Access Control (RBAC) systems to protect sensitive enterprise data.
- Designed and managed multi-tenant database schema synchronization engine across live databases.
- Developed YARP-based reverse proxy infrastructure for secure internal service routing.
- Maintained RabbitMQ-based microservices for reliable asynchronous processing.
- Optimized SQL Server queries using Dapper in high-transaction environments.
- Contributed to security modernization initiatives aligned with AFTA compliance standards.
- Automated deployment workflows with PowerShell, reducing manual errors and release time.

.NET Developer & Technical Consultant — Freelance

Remote

2020 – 2024

- Provided outsourced backend development services for SME clients, designing and delivering custom REST APIs, business automation tools, and data integration solutions.
- Consulted on software architecture and performed code reviews for early-stage startups and research-oriented projects.
- Developed cross-platform desktop and terminal applications using WPF, .NET MAUI, Blazor, Blazor WASM, and Spectre.Console (TUI) to meet specialized business requirements.
- Built internal developer tooling and CI/CD automation scripts, reducing manual deployment effort for small engineering teams.
- Delivered end-to-end technical solutions including requirement analysis, system design, implementation, testing, and production deployment.

KEY PROJECTS

Multi-Tenant Database Schema Synchronization Engine

Production system · Samarayaneh Pishro · On-demand job runner with live progress streaming

SQL SERVER SIGNALR POLLY SCHEMA DIFFING JOB ORCHESTRATION

- Built a schema extractor that introspects functions, stored procedures, table-valued parameters (TVPs), and table/column structures across tenant databases to generate a structural snapshot.
- Designed a diffing engine that compares each tenant's schema against a reference baseline and generates exact corrective migration steps.
- Enforced an additive-only change policy — the safest mode for live production data, guaranteeing sync jobs never drop or mutate existing structures.
- Built a job runner streaming real-time execution state (progress, current step, per-tenant status) via SignalR/WebSockets.
- Integrated Polly retry policies to handle transient network and SQL Server failures gracefully during execution.
- Implemented mid-run job cancellation and selective retry capabilities for failed tenants.

Enterprise Tax Platform (B2B Invoice Submission & Management)

Production system · Samarayaneh Pishro · Gateway bridge to State Tax Administration (Intamedia)

.NET CLEAN ARCHITECTURE EF CORE FLUENTVALIDATION RSA CRYPTOGRAPHY SQL SERVER XUNIT

- Refactored the core calculation engine using the Chain of Responsibility pattern, eliminating 4,000+ lines of legacy technical debt.
- Managed the full lifecycle of B2B invoices (main, cancellation, corrective, sales return) and integrated synchronization with state tax APIs (Intamedia).
- Implemented tax calculation rules and validations (VAT, duties, statutory fees) using FluentValidation.
- Secured API communication with state servers using RSA key exchange and digital signature encryption.
- Designed a comprehensive data layer (Header, Body, Payment, Shipping) with EF Core, applying Restrict Delete and Soft Delete policies.
- Implemented an InvoiceTracking logging pipeline to track request/response lifecycles and simplify debugging for REST APIs.
- Ensured zero regression in sensitive financial calculations by authoring unit and integration tests with xUnit.

Office Automation & Letter Lifecycle Management Engine (Automation Service)

Beta system · Samarayaneh Pishro · Dynamic forms, access matrix, and administrative workflow management

.NET CLEAN ARCHITECTURE RABBITMQ STATE MACHINE DYNAMIC FORM ENGINE ACCESS MATRIX EF CORE SQL SERVER

- Designed and implemented an office automation engine and letter routing pipeline powered by a custom State Machine.
- Engineered an upgraded authorization identity model (EUC) introducing USJId (User Section Job) to support multi-role organizational identities and dynamic identity switching.
- Developed a Relational Dynamic Form Builder that binds custom administrative forms to organizational positions (SectionJob).
- Implemented a multi-step approval workflow engine based on granular access levels (Approver / Signer).
- Designed an organizational Access Matrix to filter and control letter routing, receiving, workbooks, and case archives.
- Engineered Machine-to-Machine (M2M) integration with Auth Service for local cache synchronization and published file permission events to File Service via RabbitMQ.

Samateb Enterprise Healthcare Platform (Samateb Core)

Production system · Samarayaneh Pishro · Infrastructure connecting offline HIS deployments to cloud services

.NET 9 CLEAN ARCHITECTURE PARTIAL MULTI-TENANCY FINE-GRAINED RBAC YARP DAPPER EF CORE ASPIRE

- Developed a high-performance healthcare platform with .NET 9 following Clean Architecture to bridge legacy desktop HIS solutions with web apps.
- Implemented Partial Multi-Tenancy, isolating technical infrastructure configuration (ServerSettingCenter) from business settings per medical center.
- Managed dynamic database connection strings per tenant using Dapper based on domain, subdomain, and center context.
- Engineered a custom authentication and authorization pipeline using fine-grained RBAC and custom middleware for dynamic access control.
- Integrated YARP as an API Gateway for secure service routing, SSRS report proxying, and enforcing request validation rules.
- Integrated .NET Aspire and OpenTelemetry for centralized telemetry, distributed tracing, and system observability.

Medical Report Distribution & PACS Imaging System

Production system · Samarayaneh Pishro · Online diagnostic report & imaging distribution pipeline

.NET 9 DAPPER EF CORE POLLY PACS INTEGRATION REDIS SQL SERVER

- Designed a Cache-First, Tenant-Fallback report distribution engine to offload queries from on-premise HIS databases.
- Executed high-performance database queries across tenant DBs using Dapper to retrieve patient report metadata and assets.
- Integrated with multiple PACS providers (Marco, Medal, Peyvand, FilePacs) to dynamically generate viewer URLs and authentication tokens.
- Engineered a Staleness Tracking algorithm comparing UTC timestamps to automatically refresh cached report files when updated locally.
- Managed resilient asynchronous file downloads from local hospital servers using Polly retry and circuit breaker policies.
- Provided secure, anonymous access to patient reports via unpredictable GUIDs strictly scoped to tenant context headers.

CMS, Dynamic Form Builder & Multi-Tenant File Manager

Production system · Samarayaneh Pishro · Media management, public portal builder, and HIS form mapping

.NET 9 DAPPER EF CORE VIRTUAL FOLDERS QUOTA MANAGEMENT SQL SERVER

- Designed a custom healthcare CMS featuring a Drag-and-Drop Page Builder, health blog engine, sliders, and dynamic navigation menus.
- Developed a Dynamic Form Builder mapping web form submission parameters directly to legacy HIS database tables via Dapper.
- Built a multi-tenant File Manager using Virtual Folders in database metadata to eliminate expensive physical disk I/O operations.
- Implemented a dual-tier Quota Management mechanism enforcing technical file size limits and business storage quotas per tenant center.
- Optimized bulk file and folder relocation operations using specialized repository batch methods to prevent N+1 database roundtrips.
- Implemented fail-safe soft-deletion logic to prevent accidental data loss when deleting nested virtual folder structures.

Loyalty & Gamification Engine (Loyalty Club)

Production system · Samarayaneh Pishro

.NET 9 REST API SQL SERVER MEDIATR REDIS EF CORE

- Designed a comprehensive scoring and tier engine (Bronze/Silver/Gold) based on medical services, system events, and promotional occasions.
- Implemented business logic for issuing, redeeming, and validating public/private discount codes and gift packages.
- Developed a Partner Dashboard API for third-party centers (e.g., pharmacies, labs) to inquire, validate, and settle discount redemptions.
- Implemented asynchronous SMS and push notification dispatching to alert users upon tier upgrades and rewards.

Support Ticketing System

Production system · Samarayaneh Pishro

.NET 9 RATE LIMITING SIGNALR FINE-GRAINED RBAC SQL SERVER

- Developed a multi-tenant support ticketing platform connecting patients with hospital administrative staff.
- Implemented custom Rate Limiting policies to prevent spamming and abuse during ticket creation and messaging.
- Built real-time messaging workflows with advanced filtering and pagination for customer support operators.

Secure Payment Kiosk

Production system · Samarayaneh Pishro · 24/7 self-service payment terminal

WPF .NET RSA / AES CRYPTOGRAPHY PC-POS INTEGRATION QR PAYMENT

- Built a WPF desktop kiosk application for patient payments and automated admission workflows with 24/7 reliability.
- Implemented an RSA key exchange protocol with AES-encrypted communication channels to secure financial payloads.
- Delivered a fast QR-code-based payment flow integrated directly with hospital PC-POS terminals.

Enterprise Security Modernization & Technical Debt Refactoring

Samarayaneh Pishro · Legacy system upgrade and security hardening

.NET FRAMEWORK 4.8 AFTA COMPLIANCE OWASP SQL INJECTION PREVENTION CODE OBFUSCATION

- Migrated legacy enterprise applications from .NET Framework 3.5 to 4.8, patching security vulnerabilities and improving performance.
- Refactored legacy SQL queries to parameterized SQL, completely eliminating SQL injection vulnerabilities.
- Built custom developer tools for code obfuscation, digital assembly signing, and granular ACL configuration meeting AFTA security standards.

NimMas.ir — URL Shortener (Personal / Bachelor's Thesis Project)

Personal / Research project · Distributed URL shortening infrastructure

ASP.NET CORE CQRS VERTICAL SLICE REDIS DOCKER .NET ASPIRE

- Architected an ASP.NET Core web service following CQRS and Vertical Slice Architecture principles.
- Utilized Redis with dynamic TTL policies to maximize redirection speed and minimize primary database load.
- Containerized the application with Docker and orchestrated deployment using .NET Aspire on Render cloud platform.

KEY ACHIEVEMENTS

- Removed 4,000+ lines of technical debt from an enterprise tax calculation engine.
- Engineered RSA/AES-encrypted communication channels for secure payment kiosk systems.
- Built enterprise fine-grained RBAC infrastructure for sensitive financial and healthcare systems.
- Implemented a Serilog-based logging pipeline and an analytical metrics dashboard (P99, Average, Apdex, Noisy Neighbor detection).
- Automated IIS deployment, desktop build workflows, and release processes with PowerShell, including a secure reverse-proxied reporting portal.
- Contributed to healthcare software deployed across multiple medical centers and hospitals.
- Improved desktop HIS application performance by 400% through targeted SQL query optimization.

EDUCATION

Professional Software Engineering — Islamic Azad University (SAMA Andisheh)	2023 – 2026
Information Technology — Shamsipour Technical College	2020 – 2023

LANGUAGES & ADDITIONAL INFORMATION

Languages: Persian (Native), English (Professional Working Proficiency)	Soft Skills: Problem Solving, System Architecture, Cross-Functional Collaboration, Agile/Kanban, Legacy Refactoring
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